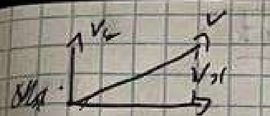

по теореме пифагора

$$V = \sqrt{V_x^2 + V_y^2}$$

$$y = v_0 \sin(a) \cdot t - \frac{gt^2}{2} \qquad y = -h$$

$$-h = v_0 \sin(a) \cdot t - \frac{gt^2}{2}$$

$$t \cdot v_0 \sin(a) - \frac{gt^2}{2} + h = 0$$

$$\cancel{v_0 \sin(a) - \frac{gt}{2} + \frac{h}{t} = 0}$$

$$\cancel{v_0 \sin(a) = \frac{gt^2 - 2h}{2t}} \qquad \cancel{v_0 \sin(a) - \frac{gt}{2} + \frac{h}{t} = 0}$$

$$\cancel{t = \frac{gt^2 - 2h}{2v_0 \sin(a)}}$$

$$t^2 \cdot \frac{g}{2} - t \cdot v_0 \sin(a) - h = 0$$

по формуле дискриминанта

$$D = v_0 \sin(a)^2 - 4\left(\frac{g}{2}\right)(-h) = (v_0 \sin(a))^2 + 2gh$$

$$t = \frac{v_0 \sin(a) + \sqrt{v_0^2 \sin(a) + 2gh}}{2 \cdot \frac{g}{2}}$$

$$t = \frac{v_0 \sin(a) + \sqrt{v_0^2 \sin(a) + 2gh}}{g}$$

$$S = V_0 \sin(a) \cdot \frac{v_0 \sin(a) + \sqrt{v_0^2 \sin(a) + 2gh}}{g}$$

$$S = \frac{v_0^2 \sin(a) \cdot \cos(a) + v_0 \cos(a) \sqrt{v_0^2 \sin^2(a) + 2gh}}{g}$$

$$v_y = v_0 \sin(a) - g\left(\frac{v_0 \sin(a) + \sqrt{v_0^2 \sin(a) + 2gh}}{g}\right)$$

$$v_y = v_0 \cancel{\sin(a)} - v_0 \cancel{\sin(a)} + \sqrt{v_0^2 \sin(a) + 2gt}$$

$$v_y = -\sqrt{v_0^2 \sin^2 a + 2gh}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = \sqrt{v_x^2 + v_0^2 \sin^2(a) + 2gh}$$

$$v = \sqrt{v_0^2 \cos^2(a) + v_0^2 \sin^2(a) + 2gh}$$

$$v = \sqrt{v_0^2((\cos^2(a) + \sin^2(a)) + 2gh}$$

$$v = \sqrt{v_0^2 + 2gh}$$

$$x = v_0 \cos a \cdot t \Rightarrow t = \frac{x}{v_0 \cos(a)}$$

$y = v_0 \sin(a) \cdot t - \frac{gt^2}{2}$ подставим t

$$y = v_0 \sin(a) \cdot \frac{x}{v_0 \cos(a)} - \frac{g\left(\frac{x}{v_0 \cos(a)}\right)^2}{2}$$

$$y = x \cdot \frac{v_0 \sin a}{v_0 \cos(a)} - \frac{g\left(\frac{x^2}{v_0^2 \cos^2(a)}\right)}{2}$$

$$y = x \cdot tg(a) - \frac{g \cdot \frac{1}{v_0^2 \cos^2(a)} \cdot x^2}{2}$$

$$y = tg(\alpha) \cdot x - \frac{g}{2 v_0^2 \cos^2(\alpha)} \cdot x^2$$